Preliminary Results of Measuring Flow Experience in a Software Modeling tool: UmpleOnline

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ABSTRACT

Weaknesses in the user experience (UX) provided by software modeling tools have been identified as important barriers reducing the uptake of such tools by developers. Emotional factors as essential parts of user experience have received little attention so far. Literature suggests that higher flow experience is associated with higher positive emotional state. Good flow experience means people feel they have clear goals and are focusing well on a task that they regard as enjoyable and are doing reasonably well at; furthermore, they do not feel a need to be concerned about time or what others are thinking and have a sense they are getting good feedback about their progress. Achieving flow is important for performance in creative tasks such as modeling. To learn more about flow we used a questionnaire-based empirical study to measure flow experience of UmpleOnline users. This paper reports preliminary results from 24 respondents, demonstrating a moderate experience of flow state in UmpleOnline. Our objective in this paper is to stimulate the research community to think about how flow can best be measured.

CCS CONCEPTS

• Human-centered computing ~ Human computer interaction (HCI)~Empirical studies in HCI

• Software and its engineering ~ Software notations and tools ~ Development frameworks and environments ~ Integrated and visual development environments

KEYWORDS

Umple, software modeling tool, user experience, flow experience

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MODELS '22 Companion, October 23–28, 2022, Montreal, QC, Canada
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ACM 978-1-4503-9467-3/22/10…$15.00
https://doi.org/10.1145/3550356.3559099

1 INTRODUCTION

Modeling is a critical activity in software engineering. It brings many benefits by presenting abstract views of complex systems and simplifying communication among stakeholders. Despite the added value of model-centered methodologies, many studies report that they are not widely used, mostly due to poor user experience (UX) of the tools [5,29,33].

Abrahão [1] defined the term *Modeling Experience* (MX) to represent the UX of modeling tools; we will use this term in the remainder of the paper. MX needs to be scoped and defined by performing ongoing theoretical and empirical studies.

We conducted a systematic literature review to investigate existing studies concerned with modeling tool evaluation [16]. The papers we analysed uncovered usability issues, utility issues (needed functionality not available), and a few other categories of MX issues such as lack of reliability, and excess cost. However, an important category of MX issues was not analysed in the existing literature: emotional factors. Many researchers addressed the importance of emotional factors in user experience by takeing different approaches such as affective design [12], Kansei engineering [23], emotional design [24], and emotional human-technology interaction [31], all highlighting users as emotional-experiencing beings. Therefore, deeply understanding emotional experiences soothes the path to improving overall UX.

One aspect of emotional experience is flow experience. As pointed out by various authors, a higher degree of flow experience is associated with higher positive emotional state of users [26], as higher positive emotional state in software engineers is directly correlated with productivity [35].

Although we assume flow experience as an indicator of emotional state, some studies correlate flow experience with other qualities of user experience, such as perceived ease of use and usefulness [3], hedonic and utility characteristics [14], flexibility and modifiability [32], and general satisfaction [27,34]. The study reported in the current paper aims to measure flow experienced by users of a software modeling tool – in other words, how well the modeling tool enables users to feel they are focused while doing productive, enjoyable work.

This study focuses on the flow experience of users of *UmpleOnline* [21] a web-based software modeling tool. It is

among the few tools that supports editing of models both textually and diagrammatically, while also providing high quality code-generation [20]. We selected this tool for investigation as it has a built-in mechanism for prompting users for feedback that any researcher can use.

2 BACKGROUND

Flow experience, which was first characterized by Csikszentmihalyi [7], is the process by which individuals engage in *autotelic* activities, activities that are so engrossing and rewarding that they are usually done for their own sake. Csikszentmihalyi [7] defined certain dimensions of flow: (1) a balance between the challenge of the task and skills of the individual, (2) a merging of action and awareness, (3) clear perceived goals, (4) unambiguous feedback, (5) focusing on the task at hand, (6) a sense of control of the activity, (7) a loss of self-consciousness or a reduced awareness of self, (8) time transformation, and (9) an autotelic, intrinsically rewarding experience. Since introducing this concept, the dimensions have been operationalized and measured variously in different domains. Different sets of dimensions are applied by many studies to assess flow experience.

Due to the clear positive effects of flow experience like increasing productivity, learning and positive experience, flow research is growing in many domains [36]. Flow experience has been assessed in areas including physical activity, education, art, gaming, and computing domains. In the latter it mostly has been studied in human-computer interaction and online-environment studies [11]. HCI (Human-Computer Interaction) Researchers use flow experience to understand human behavior in using computers and thus inform design, training and user experience [36].

Kuusinen et al. [18,19] examined flow experience in the context of developer experience using development tools: Qt Creator (An IDE) and Vaadin Designer (A graphical user interface designer tool), and their results suggest that flow experience is a significant predictor of developer experience in general. It seems therefore logical to study flow in modeling tools to improve modeling experience.

3 METHOD

The main objective is to broaden the body of knowledge in the domain of flow experience. The following objectives were set as part of this goal:

- Measuring flow experience of UmpleOnline users and examining different factors of flow experienced by users.
- Determining if there is a positive relationship between flow experience and experience level in modeling, something which has been found to be tried in other domains [6,9,22,36].
- Determining if there is any relationship between flow experience and the expression of positive or negative comments about modeling tools.

To address these objectives, we formulate our research questions as follows. The first two questions are the central ones. The third question will allow us to determine if there are connections between other MX issues and flow.

1) What do users perceive their flow to be when using UmpleOnline? Which components of flow are more experienced by users?
2) What is the relationship, if any, between flow dimensions and the modeling and professional experience level of UmpleOnline users?
3) How do respondents describe their overall MX with UmpleOnline including its strengths and weaknesses? Is there any relationship between these expressions and their flow experience?

There are several approaches and instruments to measure flow experience. Abuhamdeh [2] investigated various approaches and concluded that in majority of these studies, flow is operationalized as a continuous construct, covering the full range of participants' experience in different dimensions.

A popular method to measure flow is using direct self-report measures, because of its ease of administration [13], [25]. One of the currently most-used instruments in this category [11] is the *Short Dispositional Flow State* (SDFS-2) [15] that asks participants to indicate the extent to which the items characterize their experience in a just-completed activity on Likert scales. It measures nine dimensions of flow based on Csikszentmihalyi's definition [7]. Csikszentmihalyi [7] also highlighted "intrinsic motivation" as a key component of flow experience. Therefore we also measured intrinsic motivation in our study using some items of the *Intrinsic Motivation Indicator questionnaire* (IMI) [17].

Responses were scored on 5-point scales ranging from never to always for SDFS-2 questionnaire and strongly disagree to strongly agree for IMI questionnaire. Responses to the items on these scales are usually averaged to compute a single "flow" score for each observation [2]. For assessing general user perception of the product, we asked two open-ended questions, based on the survey questions in the literature [30].

In order to be able to categorize participants demographically, we asked about participants' country, level of education, and self-assessed level of experience in modeling. We did not ask participants to provide any identifiable characteristics to protect their privacy. In addition to demographic questions, some general questions were asked to help us better understanding of the way Umple is used by users.

To recruit participants, we followed Systematic Random Sampling [4] as our sampling method. Flow is a state of a temporary subjective experience of the human-computer interaction, that can be more accurately measured when measured immediately after an interaction [32], so we asked the UmpleOnline team to make a study invitation message appear to every 1 out of 4 users who have used it for more than 5 minutes and have done at least ten edits on a model in that time. This set of criteria ensures users have done enough modeling that is logical to ask about flow experience. The survey invitation itself does not

interrupt flow: It just appears where other types of feedback messages appear, so users can ignore it or delay their response. Users who accepted the invitation were taken to a separate tab containing a survey we created using SurveyMonkey.

4 RESULTS: BACKGROUND QUESTIONS

The survey was first made to appear in UmpleOnline in November 2021, and as of the time of writing of this paper (after 9 months), we have received 25 participants, with 24 valid answers. We screened for invalid responses by having some questions where somebody answering randomly would be inconsistent (such as saying they were both a grade-school student and a graduate student).

Ten (42%) of the valid responses were undergraduate students in university or college, 7 (25%) were practitioners in industry, 5 (21%) were graduate students in Masters or PhD programs, and 2 (8%) were educators. The majority of participants (58%) were from Canada, 15% from USA, and the others were distributed around the globe (Belgium, Brazil, Finland, France, Germany, Italy, Poland, Russia, UK). We had a relatively good distribution in terms of respondents' experience level in modeling, most of participants 42% were inexperienced, which we defined (visible to them in the question) as having created a few very simple models or having created models only in academic courses. 21% self-identified as experts meaning that they have been using modeling tools extensively in professional practice for at least 6 months. 37.5% were medium in terms of experience level, i.e. between inexperienced and expert.

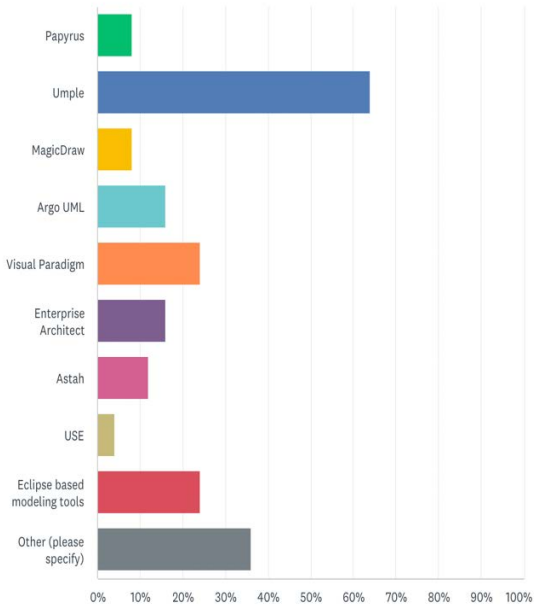


Figure 1: The modeling tools used by respondents, from Question 4 of the survey (multiple responses allowed)

Figure 1 illustrates the modeling tools that participants have used. Given that the survey was shown to UmpleOnline users, it is not surprising that Umple with 62.5% is the most frequent answer; this is followed by Visual Paradigm and Eclipse-based modeling tools. The Figure shows the tools that we explicitly listed as choices. However, there was an 'Other, please specify', option where people mentioned they had used SmartDraw, Plant UML, Umbrello, Draw.io, Lucidchart, Visio, Draw.net, Capella, PlantUML, Rhapsody, RSA, Rose, UMLET, Visual Studio, Visio Pro, and Rational Rose.

In the survey, we asked participants why they chose Umple over other modeling tools. 45% of respondents indicated that they chose Umple because it was required by their professor/supervisor/ teammate, and about half of this group also indicated that it was the only tool that they were familiar with. 11 respondents (45%) mentioned that they found Umple through a link; 4 reported that they needed one or more particular features that seem only available or are best in Umple. And Umple is introduced as participants' preferred tool by 5 users (21%).

We asked participants which Umple tools they have used. As expected, all participants selected UmpleOnline as one of the tools, but 29% also selected IDE Umple Plugins for Eclipse or VSCode, and 12.5% indicated they had used the Umple command-line compiler.

We also asked participants what is the main task that they use Umple for. The answers were formed as four pre-defined options, with an "Other" option to enable respondents to answer freely. 67% used Umple for drawing diagrams, 17% used it for generating a small functioning system, 8% used it to build a large complex system, and 4% for analyzing a model for errors and weaknesses. 4% mentioned that they use Umple for all the above tasks. There may be several reasons for most people only using the tool for diagramming: it could be due to features like analysis or code generation being less known or not required for the user's goals.

5 RESULTS: FLOW QUESTIONS

As discussed earlier, to capture users flow experience, we employed the SDFS-2 and IMI questions. Using these, we measured each construct of flow with one measurement. One of the most-cited factors contributing to flow experience is the balance of *challenge* perceived by users and the *skill* that they have in the task [6,9,22,36]. The high challenge inherent in software engineering has been documented as contributing to developer motivation [10].

We calculated mean, standard deviation, the percentage of respondents who gave the two top-most Likert scale responses (agree and strongly agree, also scored 4 and 5 in our other calculations), as well the percentage who responded with the bottom-two responses (disagree and strongly disagree, scored 2 and 1). We summarize results for each subscale in Table 1.

All mean scores were above the neutral point (>3.0), with a mean of **3.42** which could be interpreted as a feeling of a moderate state of flow in respondents. The flow dimension with

the most respondents saying they agree or strongly agree, is **Concentration on task (82.6%),** followed by **Loss of self-consciousness (72.7%).** On the other hand, **Action awareness** and **Transformation of time** are the flow dimensions with highest percentages of respondents choosing to disagree and strongly disagree, although in both cases the agreement still exceeded the disagreement.

Table 1. Results of SDFS-2 subscales, measuring agreement on a scale of 1-5 with various flow statements

Flow dimensions	SDFS-2 item	Mean	St. Dev.	% Top 2	% Bottom 2
Q8 Challenge-skill	I feel I am competent enough to meet the high demands of the situation	3.46	1.10	58.3	16.7
Q9 Action awareness	I do things spontaneously and automatically without having to think	3.13	1.01	43.5	30.4
Q10 Clear goals	I have a strong sense of what I want to do	3.50	1.10	62.5	16.7
Q11 Unambiguous feedback	I have a good idea while I am performing about how well I am doing	3.52	1.04	60.9	17.4
Q12 Concentration on task	I am completely focused on the task at hand	3.91	0.85	82.6	4.3
Q13 Sense of control	I have a feeling of total control	3.43	0.99	56.5	17.4
Q14 Loss of self-consciousness	I am not worried about what others may be thinking of me	3.91	1.06	72.7	9.1
Q15 Transformation of time	The way time passes seems to be different from normal	3.00	1.22	42.9	28.6
Q16 Autotelic experience	The experience is extremely rewarding	3.22	0.95	43.5	21.7

We also utilized some items from the IMI questionnaire to measure *intrinsic motivation* as an effective factor of flow experience. Intrinsic motivation is often contrasted with *extrinsic* motivation in which the incentive resides in the expected external results rather than the activity itself [28]. We selected the following IMI subscales, similar to [9]: interest/enjoyment, perceived competence, effort/importance, and perceived choice, the results are summarized in Table 2.

There is disagreement regarding the "I have no choice" statement (which is deliberately expressed in an inverse sense), indicating that participants felt they *do indeed have* choice; this is promising, since tool choice might naturally be sometimes limited by workplace or academic demands (i.e. the team is using the tool, or the professor assigns it.)

Table 2. Results of IMI subscales, measuring agreement on a scale of 1-5 with aspects of motivation

IM dimensions	IMI item	Mean	St. Dev	% Top 2%	% Bottom 2
Q17 Interest / enjoyment	I enjoy using Umple very much.	3.67	0.92	66.7	8.3
Q18 Perceived competence	I am pretty skilled in using Umple.	2.79	0.98	16.7	41.7
Q19 Perceived choice	I use Umple because I have no choice. [inversely phrased]	2.58	0.97	16.7	54.2
Q20 Effort / importance	It is important to me to do well while using Umple.	3.46	1.06	54.2	12.5

6 RESULTS: RELATIONSHIP BETWEEN EXPERIENCE AND FLOW

To investigate our research question about the relationship between experience and flow, we tested correlation between experience level of respondents and different dimensions of both flow experience and intrinsic motivation. As the variables were continuous and quantitative, we used Pearson's correlation to examine the linear association between these variables.

The results showed a moderate correlation (0.57) between experience level and competency level. Several researchers including Csikszentmihalyi [8] believe that "the state of flow is felt when opportunities for action are in balance with the actor's skills" Csikszentmihalyi and his colleagues sometimes operationalized flow based solely on the ratio of challenges and skills [2]. Thus we can conclude that flow experience is more likely to occur when an individual has a good level of experience and skill in the domain. This result is aligned with literature claiming that flow is more likely to be experienced by those who have developed skills in the task at hand [6,9,22,36].

The correlation test also showed negative correlation between experience level and perceived choice (-0.46), stating that more experienced users perceive more sense of choice in comparison to inexperienced users. This result also makes sense since a more experienced user is more likely to have prior experience using different modeling tools or not to be bound by a manager/professor's dictates, and therefore they chose the current tool based on their preference rather than having no other choices.

7 RESULTS: FREE-FORM QUESTIONS

There were 8 responses to free-form questions asking about the best qualities of UmpleOnline and areas in need of improvement. Four respondents pointed to "Code generation" and the high quality of it as the best quality of UmpleOnline. Some features mentioned by respondents are including text/diagram duality, simplicity, clean syntax, free and open source, well supported, exporting options, supporting both beginners and experienced users to use the tool effectively.

Respondents consider better UI (User Interface) and UX as qualities that need improvement in UmpleOnline. Cleaner generated code, better graphical support of big models, providing more information about how the Umple compiler works, adding a button to manually re-render diagrams, and supporting C# are among the points that are mentioned by respondents to improve Umple.

Unfortunately, few respondents with the lowest rate of flow answered these two questions, so drawing a meaningful relationship among the flow factors and improvements suggested by users was not possible. This is a work-in-progress, so we hope that by receiving more responses, we will be able to draw more insightful conclusions.

8 THREATS TO VALIDITY AND LIMITATIONS

The main threat to the validity of this paper is internal validity, arising from the small sample size of participants, we tried to mitigate this threat by randomizing participant recruitment from a large pool. We solicit 1 out of every 4 users who meet our criteria for time using UmpleOnline and edits performed in their session. A total of 2190 people were invited to participate, of which only 77 (3.5%) clicked the link, and only 24 submitted valid responses (1.1%). Since so many websites now ask people to do surveys, we attribute this low response rate simply to survey response fatigue. It is also possible that people who are "in the zone", i.e. with a good sense of flow, are skipping our survey because they want to continue with their task; alternatively it could be that only the most enthusiastic or most negative-feeling people are more motivated to respond. Any of these factors could be providing a bias.

It does seem that participants were well-distributed among education levels, occupations, tool experience and similar factors therefore reducing demographic bias.

As this is a work-in-progress paper, we hope that we could receive more responses soon, leading to a higher internal validity. We will consider changing the sampling and recruitment strategy to reach this goal.

There is a threat to external validity, since the survey only applies to one modeling tool, UmpleOnline. We plan to extend our work and include other software modeling tools to strengthen the external validity of results.

To mitigate the threat to construct validity of this paper, the majority of survey questions come from validated surveys so the measures and constructs are reliable. We also performed a pilot study to identify any flaws and misunderstandings of our questions. Moreover, we considered "other" options in answers items whenever it was applicable, to include all possible answers.

9 CONCLUSION

In this study, we conducted a survey to investigate flow experience of UmpleOnline users. We collected and analysed 24 responses. Respondents were from both industry and academia, with different levels of expertise in modeling and from different countries, so we expect that this set of respondents formed a representative sample of our target group of users.

To measure flow experience of participants, we used SDFS-2 with 9 dimensions of flow and IMI with 4 dimensions, and calculated average score of each subscale, and then an overall flow score. The results showed an intermediate level of flow experience in almost all factors. In this study we disregarded the mediator factors, but based on the research [26,36] there are several contextual factors like cultural and personal traits e.g 'autotelic personality', mediating experience of flow. Further research is needed to examine and assess these factors in the software modeling field.

At a later date it would be good to compare the flow experience of users of other modeling tools, and also Umple IDE and Umple command line users.

This study served as a basis to consider emotional factors in software modeling tools. We are motivated to expand this research by including other modeling tools to enable us to compare them in terms of flow experience, and we also encourage other researchers to build upon our results and contribute further knowledge in the domain of flow experience in the domain of software modeling.

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